

SINGLE-QUBIT GATES

Pauli gates, Hadamard interference, phase operations, continuous rotations, Bloch-sphere geometry, decompositions, and universal gate synthesis

A detailed reference for quantum computing, quantum information, QKD, algorithms, and circuit design.

Topic 11 in the Quantum Mathematics Foundation series

How to use this volume

Single-qubit gates are the smallest complete laboratory for quantum control. Every gate is a 2×2 unitary matrix, yet this compact setting already contains complex amplitudes, interference, noncommuting operations, continuous rotations, basis changes, global and relative phase, and universal synthesis. The chapter develops each standard gate algebraically and geometrically, then connects the gates through rotation theory and decomposition.

Learning objectives

- Apply common one-qubit gates to arbitrary state vectors.
- Interpret Pauli, phase, Hadamard, and rotation gates on the Bloch sphere.
- Distinguish population changes, relative-phase changes, and global phase.
- Compose gates in the correct order and recognize noncommutativity.
- Derive R_x , R_y , R_z and arbitrary-axis rotations from matrix exponentials.
- Express a general one-qubit unitary using Euler rotations and global phase.
- Prepare an arbitrary pure qubit state from $|0\rangle$.
- Understand Clifford gates, the role of T , and common circuit identities.

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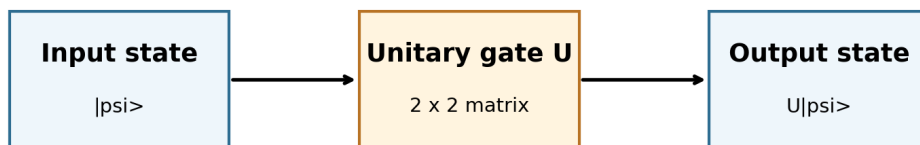
1. What a single-qubit gate is

A pure one-qubit state is a normalized vector in $\mathbb{C}^2$. A closed-system gate must transform every normalized state into another normalized state, preserve inner products, and be reversible. These requirements make the gate a 2×2 unitary matrix U .

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = [\alpha, \beta]^T, |\alpha|^2 + |\beta|^2 = 1.$$

$$|\psi_{\text{out}}\rangle = U|\psi_{\text{in}}\rangle, U^\dagger U = I.$$

A single-qubit gate is a reversible linear transformation of $\mathbb{C}^2$



The gate changes amplitudes while preserving total probability.

The word gate describes both the abstract unitary operation and its implementation in a circuit. A physical pulse, an optical element, or a compiled sequence may realize the same mathematical unitary.

2. State vectors and matrix action

Write a general matrix as

$$U = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix}.$$

Acting on $|\psi\rangle$ gives

$$U[\alpha, \beta]^T = [u_{00}\alpha + u_{01}\beta, u_{10}\alpha + u_{11}\beta]^T.$$

The first column is $U|0\rangle$; the second column is $U|1\rangle$. Therefore a gate is completely determined by its action on the computational basis.

$$U|\psi\rangle = \alpha U|0\rangle + \beta U|1\rangle.$$

This is the operational meaning of linearity. The gate acts independently on the basis branches and then the resulting amplitudes add coherently.

A compact atlas of common single-qubit gates

I

$|0\rangle \rightarrow |0\rangle$
 $|1\rangle \rightarrow |1\rangle$

X

$|0\rangle \rightarrow |1\rangle$
 $|1\rangle \rightarrow |0\rangle$

Y

$|0\rangle \rightarrow i|1\rangle$
 $|1\rangle \rightarrow -i|0\rangle$

Z

$|0\rangle \rightarrow |0\rangle$
 $|1\rangle \rightarrow -|1\rangle$

H

$|0\rangle \rightarrow |+\rangle$
 $|1\rangle \rightarrow |-\rangle$

S / T

$|0\rangle$ unchanged
phase on $|1\rangle$

3. Unitarity and reversibility

Unitarity means

$$U^\dagger U = U U^\dagger = I, U^{-1} = U^\dagger.$$

For any vectors $|\phi\rangle$ and $|\psi\rangle$,

$$\langle U\phi | U\psi \rangle = \langle \phi | U^\dagger U | \psi \rangle = \langle \phi | \psi \rangle.$$

Consequences

- Norms remain one, so total probability is preserved.
- Orthogonal states remain orthogonal, so distinguishability is preserved.
- Angles and transition amplitudes are preserved.
- The operation can be exactly undone by $U^\dagger$.
- The columns and rows of U form orthonormal bases.

A matrix that shrinks every state, deletes one component, or maps two distinct states to one state cannot be a closed-system quantum gate. Measurement and noise can be nonunitary because they involve classical outcomes, discarded environments, or conditioning.

Quick column test

For a 2×2 matrix, check that both columns have norm one and their inner product is zero. If so, the matrix is unitary.

4. Identity and global-phase gates

The identity gate is

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, I|\psi\rangle = |\psi\rangle.$$

A global-phase gate is

$$G(\delta) = e^{i\delta} I.$$

It maps every state to $e^{i\delta}|\psi\rangle$. All amplitudes rotate by the same complex phase, so measurement probabilities and all later closed-system predictions remain unchanged.

Physical equivalence

$$|\psi\rangle \sim e^{i\delta}|\psi\rangle.$$

The matrices U and $e^{i\delta}U$ can therefore implement the same projective action on a single isolated state, although relative phases can become meaningful when a unitary appears as one branch of a larger controlled operation.

Important subtlety

A phase that is global for a complete state may be relative inside a larger superposition. Controlled- U can make the global phase of U observable relative to the branch on which U was not applied. This is why circuit synthesis sometimes tracks global phase even when statevector comparison ignores it.

5. Pauli-X gate

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Its basis action is

$$X|0\rangle = |1\rangle, X|1\rangle = |0\rangle.$$

For a general state,

$$X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle.$$

Different gates modify different parts of the quantum state

Pauli-X swaps amplitudes

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \begin{bmatrix} \beta \\ \alpha \end{bmatrix}$$

Population exchange

Pauli-Z changes relative phase

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \begin{bmatrix} \alpha \\ -\beta \end{bmatrix}$$

Probabilities unchanged in Z basis

Bloch-sphere action

X is a 180-degree rotation about the x axis, up to a global phase when compared with the conventional rotation operator $R_x(\pi)$. It maps Bloch coordinates

$$(x, y, z) \rightarrow (x, -y, -z).$$

Useful identities

$$X^2 = I, X^\dagger = X, X^{-1} = X.$$

X is both unitary and Hermitian. It is a gate and also an observable with eigenstates $|+\rangle$ and $|-\rangle$ and eigenvalues +1 and -1.

6. Pauli-Z gate

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Its basis action is

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

For a general state,

$$Z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle.$$

The computational-basis probabilities are unchanged, but the relative phase changes by π . This phase difference can later become a population difference through interference.

Bloch-sphere action

Z is a 180-degree rotation about the z axis and maps

$$(x, y, z) \rightarrow (-x, -y, z).$$

Example

$$Z|+\rangle = |-\rangle, \quad Z|-\rangle = |+\rangle.$$

As with X,

$$Z^2 = I, \quad Z^\dagger = Z, \quad Z^{-1} = Z.$$

7. Pauli-Y gate

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}.$$

Its basis action is

$$Y|0\rangle = i|1\rangle, Y|1\rangle = -i|0\rangle.$$

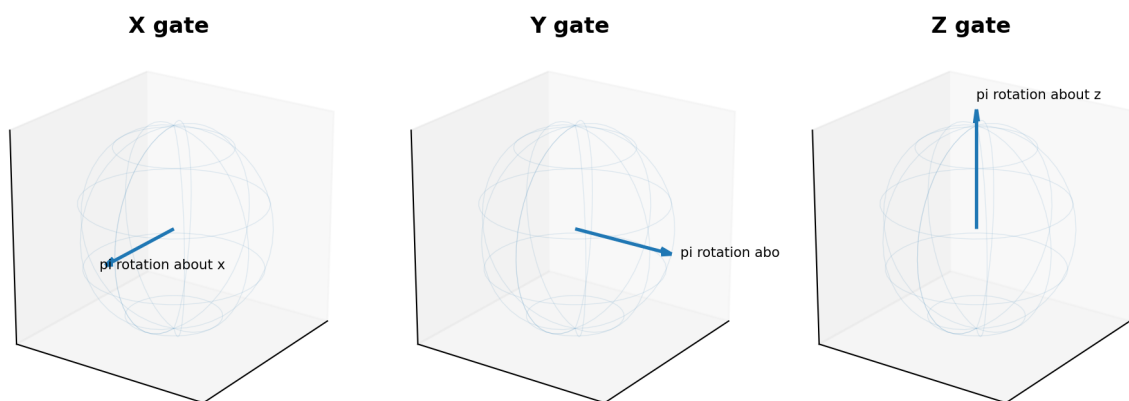
For a general state,

$$Y[\alpha, \beta]^T = [-i\beta, i\alpha]^T.$$

Y combines a bit exchange with phase factors. Algebraically,

$$Y = iXZ = -iZX.$$

Pauli gates are 180-degree Bloch-sphere rotations



Bloch-sphere action

Y is a 180-degree rotation about the y axis and maps

$$(x, y, z) \rightarrow (-x, y, -z).$$

The eigenstates are

$$|+i\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}, \quad |-i\rangle = (|0\rangle - i|1\rangle)/\sqrt{2}.$$

8. Pauli algebra

The Pauli matrices satisfy

$$X^2=Y^2=Z^2=I.$$

$$XY=iZ, YZ=iX, ZX=iY.$$

Reversing the order changes the sign:

$$YX=-iZ, ZY=-iX, XZ=-iY.$$

Pauli multiplication is noncommutative and phase-sensitive

	I	X	Y	Z
I	I	X	Y	Z
X	X	I	iZ	-iY
Y	Y	-iZ	I	iX
Z	Z	iY	-iX	I

Example: $XY = iZ$, while $YX = -iZ$.

Commutators and anticommutators

$$[X,Y]=XY-YX=2iZ, \{X,Y\}=XY+YX=0,$$

with cyclic permutations. More compactly,

$$\sigma_j \sigma_k = \delta_{jk} I + i \sum_l \epsilon_{jkl} \sigma_l.$$

Every 2×2 complex matrix can be expanded in the operator basis $\{I, X, Y, Z\}$. Every Hermitian one-qubit observable has real coefficients in this basis.

9. Hadamard gate

$$H = (1/\sqrt{2})[[1,1],[1,-1]].$$

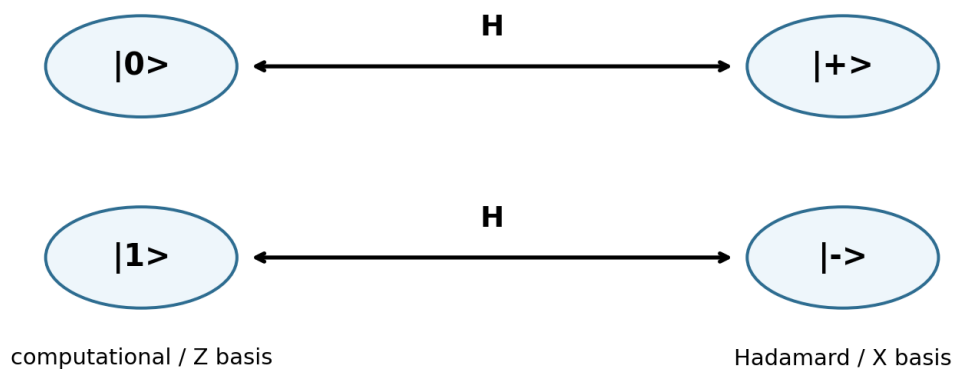
Its basis action is

$$H|0\rangle = |+\rangle, H|1\rangle = |-\rangle.$$

The reverse relations also hold because $H^2=I$:

$$H|+\rangle = |0\rangle, H|-\rangle = |1\rangle.$$

Hadamard connects the Z and X eigenbases



General state action

$$H[\alpha, \beta]^T = [(\alpha + \beta)/\sqrt{2}, (\alpha - \beta)/\sqrt{2}]^T.$$

Hadamard does not simply create randomness. It changes basis and recombines amplitudes. Depending on the input phase, its output may be deterministic or probabilistic.

Bloch geometry

On Bloch vectors, H exchanges the x and z axes and reverses y:

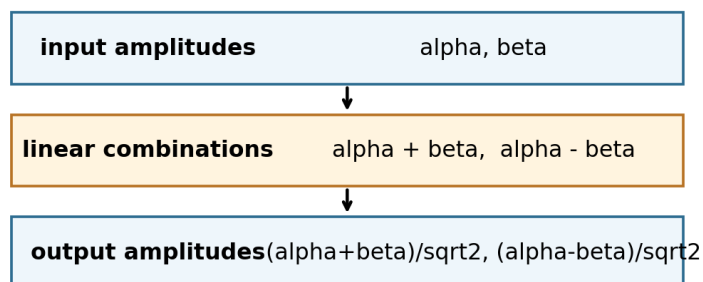
$$(x, y, z) \rightarrow (z, -y, x).$$

10. Hadamard and interference

Hadamard performs a sum and a difference of amplitudes:

$$\alpha_{\text{out}} = (\alpha + \beta) / \sqrt{2}, \quad \beta_{\text{out}} = (\alpha - \beta) / \sqrt{2}.$$

Hadamard turns phase relationships into amplitude differences



Equal phases reinforce one output; opposite phases reinforce the other.

Constructive and destructive interference

- If $\alpha = \beta$, the lower output cancels and the upper output reinforces.
- If $\alpha = -\beta$, the upper output cancels and the lower output reinforces.
- If the relative phase is neither 0 nor π , interference is partial.

For the equatorial state

$$|\psi_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle) / \sqrt{2},$$

a Hadamard followed by Z measurement gives

$$P(0) = \cos^2(\phi/2), \quad P(1) = \sin^2(\phi/2).$$

This phase-to-probability conversion is a recurring mechanism in quantum algorithms.

11. Phase gate family $P(\lambda)$

The general phase gate is

$$P(\lambda) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\lambda} \end{bmatrix}.$$

It acts as

$$P(\lambda)(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle + e^{i\lambda}\beta|1\rangle.$$

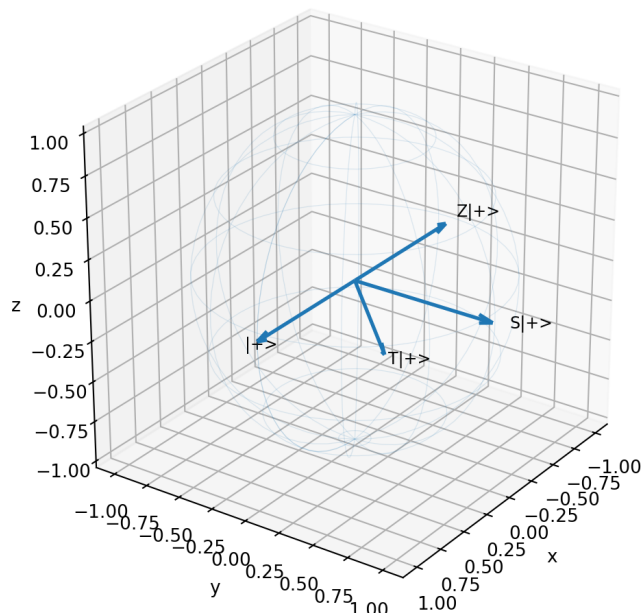
It preserves computational-basis probabilities and adds λ to the relative phase of $|1\rangle$ with respect to $|0\rangle$.

Relation to z rotation

$$P(\lambda) = e^{i\lambda/2} R_z(\lambda).$$

Thus $P(\lambda)$ and $R_z(\lambda)$ have the same Bloch-sphere action but differ by a global phase.

Phase gates rotate states around the Bloch z-axis



Composition

$$P(\lambda_1)P(\lambda_2) = P(\lambda_1 + \lambda_2).$$

12. S and T gates

The S gate is a quarter-turn phase gate:

$$S = P(\pi/2) = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, S^2 = Z, S^4 = I.$$

The T gate is an eighth-turn phase gate:

$$T = P(\pi/4) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}, T^2 = S, T^4 = Z, T^8 = I.$$

Adjoins

$$S^\dagger = P(-\pi/2), T^\dagger = P(-\pi/4).$$

Quantum role

- S is a Clifford gate. It maps Pauli operators to Pauli operators under conjugation.
- T is non-Clifford. Adding T to Clifford operations enables approximate universal quantum computation.
- Both gates change phase without changing Z-basis populations.

Example

$$S|+\rangle = (|0\rangle + i|1\rangle)/\sqrt{2} = |+i\rangle.$$

13. Rotation gates Rx, Ry, Rz

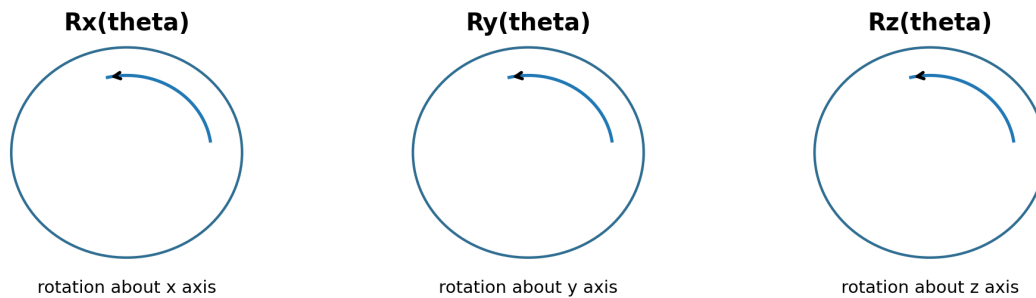
Continuous one-qubit control is expressed through axis rotations:

$$R_x(\theta) = \exp(-i \theta X/2) = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}.$$

$$R_y(\theta) = \exp(-i \theta Y/2) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}.$$

$$R_z(\theta) = \exp(-i \theta Z/2) = \begin{bmatrix} e^{-i \theta/2} & 0 \\ 0 & e^{i \theta/2} \end{bmatrix}.$$

Parameterized gates provide continuous Bloch-sphere rotations



The half-angle is fundamental. Statevectors are spinors: a Bloch-vector rotation by θ is represented by matrix amplitudes containing $\theta/2$.

Special values

$$R_x(\pi) = -iX, R_y(\pi) = -iY, R_z(\pi) = -iZ.$$

The extra factor $-i$ is global for an isolated qubit state.

14. Rotations from exponentials

For any Pauli matrix σ with $\sigma^2=I$, expand the exponential:

$$\exp(-i \theta \sigma / 2) = \sum_n [(-i \theta / 2)^n \sigma^n / n!].$$

Even powers contribute I and odd powers contribute σ . Grouping the series gives

$$R_\sigma(\theta) = \cos(\theta/2)I - i \sin(\theta/2)\sigma.$$

This single identity produces R_x , R_y , and R_z .

Composition on the same axis

$$R_k(a)R_k(b)=R_k(a+b), \quad R_k(\theta)^\dagger=R_k(-\theta).$$

Periodicity

$$R_k(\theta+2\pi)=-R_k(\theta), \quad R_k(\theta+4\pi)=R_k(\theta).$$

The sign after 2π is a global phase for an isolated state, while the exact matrix returns only after 4π . This is the $SU(2)$ spinor structure behind qubit rotations.

15. Arbitrary-axis rotations

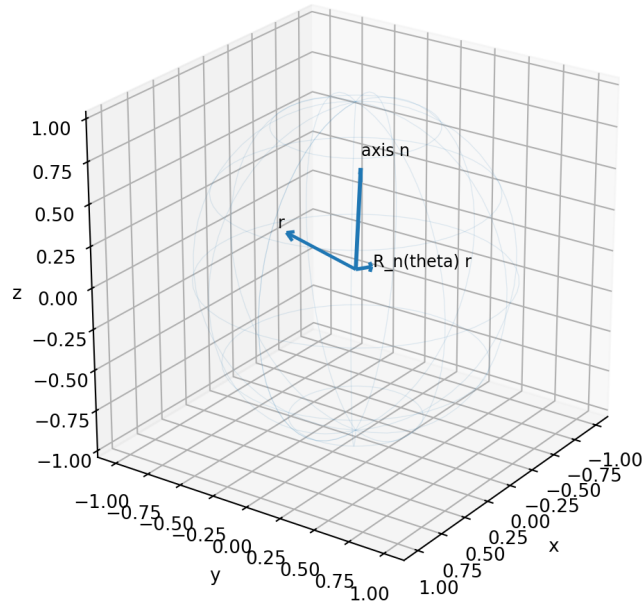
Let $n=(n_x, n_y, n_z)$ be a real unit vector. Define

$$n \cdot \sigma = n_x X + n_y Y + n_z Z.$$

Because $(n \cdot \sigma)^2 = I$, the rotation about axis n is

$$R_n(\theta) = \exp[-i \theta (n \cdot \sigma)/2] = \cos(\theta/2) I - i \sin(\theta/2) (n \cdot \sigma).$$

General one-qubit rotations use an arbitrary Bloch axis



On the Bloch sphere, the Bloch vector r transforms by the ordinary three-dimensional rotation of r about n through angle θ . The statevector matrix belongs to $SU(2)$, while the Bloch-vector rotation belongs to $SO(3)$.

Generator interpretation

The Hermitian operator $(n \cdot \sigma)/2$ generates the continuous rotation. For very small $d\theta$,

$$R_n(d\theta) \text{ approximately } I - i(d\theta/2)(n \cdot \sigma).$$

16. Bloch-sphere action

A normalized pure qubit can be written as

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle.$$

Its Bloch vector is

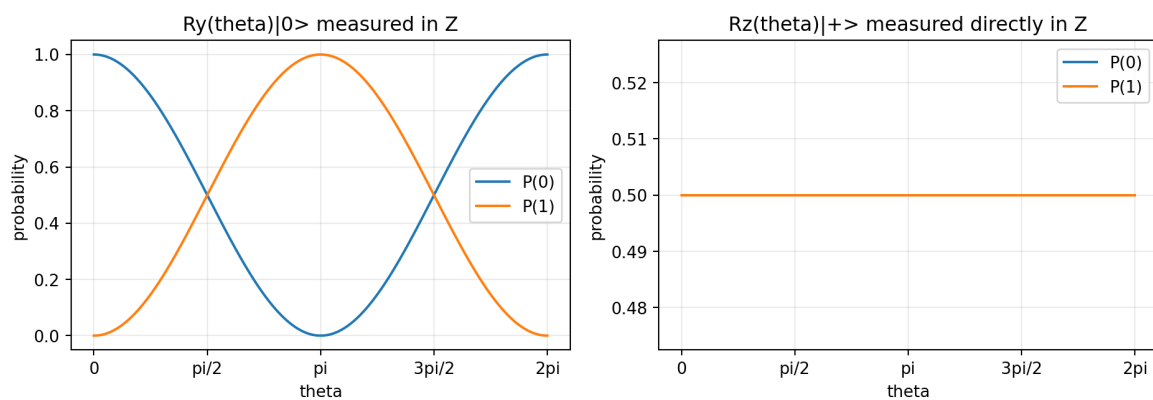
$$\mathbf{r} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta).$$

Unitary conjugation transforms the density matrix

$$\rho = (\mathbf{I} + \mathbf{r} \cdot \boldsymbol{\sigma})/2 \rightarrow U \rho U^\dagger = (\mathbf{I} + \mathbf{r}' \cdot \boldsymbol{\sigma})/2.$$

For a one-qubit unitary, $\mathbf{r}'$ is a rigid rotation of $\mathbf{r}$. Norm and purity are preserved.

Rotations can change populations or only relative phase



Population versus phase

R_y commonly changes Z-basis populations because it moves the state north or south. R_z changes azimuth and relative phase but leaves Z-basis populations unchanged until a later basis-changing gate exposes that phase.

17. Gate composition and order

If U_1 acts first and U_2 acts second, the total state is

$$|\psi_{\text{out}}\rangle = U_2 U_1 |\psi_{\text{in}}\rangle.$$

The rightmost matrix acts first. Matrix multiplication is generally noncommutative:

$$U_2 U_1 \neq U_1 U_2.$$

Gate order matters: the rightmost matrix acts first

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{Z} |-\rangle$$

$$\mathbf{ZH}|0\rangle = |-\rangle$$

$$|0\rangle \xrightarrow{Z} |0\rangle \xrightarrow{H} |+\rangle$$

$$\mathbf{HZ}|0\rangle = |+\rangle$$

Therefore ZH and HZ are different transformations.

Example

$$\mathbf{ZH}|0\rangle = |-\rangle, \mathbf{HZ}|0\rangle = |+\rangle.$$

Circuit reading

Circuit diagrams are read from left to right in time, but algebraic products place later gates to the left. A circuit U_1 then U_2 then U_3 corresponds to $U_3 U_2 U_1$.

Same-axis rotations commute, but rotations about different axes usually do not. For small angles, the commutator controls the leading order difference.

18. Conjugation of observables

The expression $U A U^\dagger$ describes how an observable or Bloch axis is rotated by U . Equivalently, $U^\dagger A U$ is the observable seen in the transformed measurement frame.

Conjugation describes how a gate transforms observables and axes

$H Z H$	=	X	Hadamard exchanges x and z axes
$H X H$	=	Z	same transformation in reverse
$S X S^\dagger$	=	Y	quarter-turn about z
$X Z X$	=	$-Z$	pi-turn flips the z direction

Important identities

$$H X H = Z, H Z H = X, H Y H = -Y.$$

$$S X S^\dagger = Y, S Y S^\dagger = -X, S Z S^\dagger = Z.$$

$$X Z X = -Z, Z X Z = -X.$$

Why this matters

- Basis-change measurements are derived by conjugating observables.
- Clifford circuits can be tracked through their action on Pauli operators.
- Error propagation through gates is often analyzed by conjugation.
- Rotation identities can be checked by examining how axes transform.

19. General single-qubit unitary

A general 2×2 unitary has four real degrees of freedom. One is a global phase, leaving three physically relevant rotation parameters.

A useful parameterization is

$$U(\theta, \phi, \lambda) = \begin{bmatrix} \cos(\theta/2) & -e^{i\lambda}\sin(\theta/2) \\ e^{i\phi}\sin(\theta/2) & e^{i(\phi+\lambda)}\cos(\theta/2) \end{bmatrix}.$$

Up to a convention-dependent global phase, this is equivalent to

$$R_z(\phi) R_y(\theta) R_z(\lambda).$$

Parameter roles

Parameter	Geometric role
θ	polar rotation controlling amplitude mixing
ϕ	output azimuth / phase of the lower component
λ	input-relative phase convention / final z rotation
global δ	common phase multiplying the full matrix

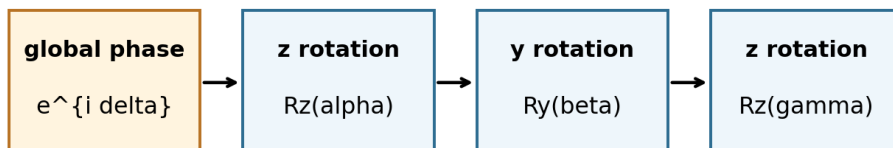
Different software libraries use different names and phase conventions. Always verify the exact matrix definition rather than relying only on a gate label.

20. Euler-angle decomposition

Every one-qubit unitary can be written as a global phase times three elementary rotations. A standard Z-Y-Z form is

$$U = e^{i\delta} R_z(\alpha) R_y(\beta) R_z(\gamma).$$

Every single-qubit unitary is rotations plus a global phase



One common convention: $U = e^{i\delta} R_z(\alpha) R_y(\beta) R_z(\gamma)$.

Why three rotations?

A physical pure-state direction has two angular degrees of freedom, but a full unitary also determines how every direction and phase relation transforms. Ignoring global phase, $SU(2)$ has three real degrees of freedom.

Other valid decompositions

$$U = e^{i\delta} R_z(a) R_x(b) R_z(c),$$

$$U = e^{i\delta} R_x(a) R_y(b) R_x(c).$$

Hardware compilers choose decompositions that fit native control axes, calibration quality, and virtual phase updates.

21. Arbitrary state preparation

Every normalized pure qubit, up to global phase, has the form

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle.$$

Starting from $|0\rangle$, one convenient preparation is

$$|\psi\rangle = R_z(\phi) R_y(\theta)|0\rangle \text{ up to a global phase.}$$

Preparing an arbitrary pure qubit from $|0\rangle$



Up to global phase, two rotation angles specify every pure one-qubit state.

Derivation

$$R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle.$$

Then $R_z(\phi)$ applies opposite half-phases. After removing the common factor $e^{-i\phi/2}$, the relative phase becomes $e^{i\phi}$.

Preparation from amplitudes

For a normalized state $\alpha|0\rangle + \beta|1\rangle$, choose $\theta = 2 \arccos(|\alpha|)$ and relative phase $\phi = \arg(\beta) - \arg(\alpha)$. The phase $\arg(\alpha)$ is global and can be omitted.

22. Clifford gates

The single-qubit Clifford group consists of gates that map Pauli operators to Pauli operators, up to signs, under conjugation:

$$C P C^\dagger \text{ belongs to } \{+/-X, +/-Y, +/-Z\}.$$

It is generated by H and S. Geometrically, single-qubit Cliffords are the orientation-preserving symmetries of the Bloch sphere that permute the x, y, and z axes.

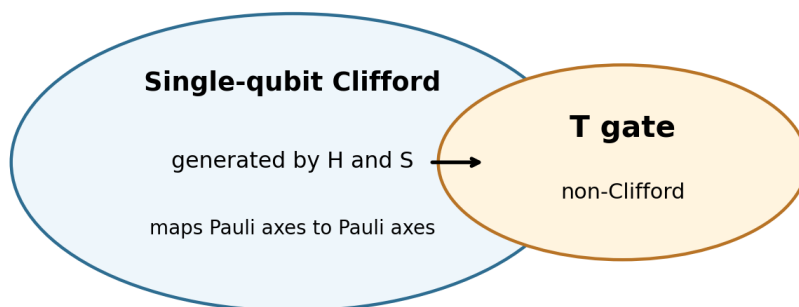
Examples

- H exchanges X and Z and flips Y.
- S maps X to Y, Y to -X, and leaves Z unchanged.
- Pauli gates themselves are Clifford gates.

Why Cliffords are special

- They preserve the Pauli operator structure.
- Stabilizer states remain stabilizer states.
- Clifford-only circuits admit efficient classical simulation in the stabilizer formalism.
- They are central to error correction, randomized benchmarking, and circuit compilation.

Clifford gates are easy to structure; T adds non-Clifford power



Clifford + T forms a standard approximately universal gate set.

23. Non-Clifford T and universality

The T gate rotates about z by $\pi/4$ up to global phase. It does not map every Pauli axis exactly to another Pauli axis, so it lies outside the Clifford group.

Universal synthesis

The set consisting of Clifford operations plus T can approximate arbitrary unitaries to any desired accuracy when combined with an entangling two-qubit Clifford such as CNOT. For single-qubit synthesis, H and T already generate a dense subset of $SU(2)$.

Why T is costly

In many fault-tolerant architectures, Clifford operations are comparatively easy while high-fidelity non-Clifford resources require magic-state preparation and distillation. Circuit cost is therefore often measured by T count and T depth.

Approximation

An arbitrary continuous rotation generally cannot be represented exactly by a finite Clifford+T sequence. Compilation selects a sequence whose unitary is within a chosen error tolerance of the target.

Do not confuse universality with speed

A gate set may be universal in principle yet inefficient for a specific hardware platform or algorithm. Native gates and compilation overhead determine practical performance.

24. Useful gate identities

Identity	Meaning
$H^2 = X^2 = Y^2 = Z^2 = I$	self-inverse gates
$S^2 = Z, T^2 = S$	phase hierarchy
$HXH = Z, HZH = X$	Hadamard exchanges X and Z
$SXS^\dagger = Y$	S rotates the x axis to y
$R_k(a)R_k(b) = R_k(a+b)$	same-axis rotations add
$R_k(\theta)^\dagger = R_k(-\theta)$	inverse rotation
$P(\lambda) = e^{i\lambda/2} R_z(\lambda)$	same physical Bloch rotation
$X R_x(\theta) X = R_x(\theta)$	same-axis conjugation
$X R_z(\theta) X = R_z(-\theta)$	perpendicular Pauli reverses rotation
$H R_z(\theta) H = R_x(\theta)$	basis-transformed rotation

Identity verification strategies

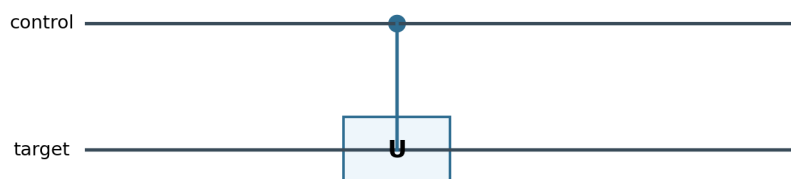
- 1 Multiply matrices directly.
- 2 Check action on a complete basis.
- 3 Compare Bloch-axis transformations.
- 4 Use exponential conjugation: $U e^A U^\dagger = e^{\{U A U^\dagger\}}$.
- 5 Allow for global phase when comparing physical state action, but not when exact controlled behavior matters.

25. Controlled versions and larger circuits

A controlled-U applies U to a target qubit only when a control qubit is $|1\rangle$. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, its block matrix is

$$C(U) = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix}.$$

A controlled-U uses a one-qubit gate conditionally inside a larger system



The target gate U remains a 2×2 single-qubit unitary, but the complete controlled operation is 4×4 .

Why global phase becomes subtle

Replacing U by $e^{i\delta}U$ changes only the control=1 branch. That phase is now relative to the control=0 branch and can affect interference. Therefore controlled- U and controlled- $(e^{i\delta}U)$ are not generally equivalent.

Examples

- Controlled-X is CNOT.
- Controlled-Z applies a -1 phase only to $|11\rangle$.
- Controlled phase $P(\lambda)$ applies $e^{i\lambda}$ only to $|11\rangle$.
- Controlled rotations appear in phase estimation, simulation, and variational circuits.

Although U is a one-qubit gate, the controlled operation acts on a four-dimensional two-qubit space and can create entanglement.

26. Physical implementation intuition

The same mathematical gate can be implemented differently across technologies.

Platform	Typical control picture
Superconducting qubit	microwave pulses drive rotations in the equatorial plane; frame updates implement z rotations efficiently
Trapped ion	laser interactions control internal states; phases and pulse areas set rotation axes and angles
Photonic qubit	wave plates, interferometers, and phase shifters implement polarization or path unitaries
Spin qubit	magnetic or electric control drives spin resonance and phase accumulation
Neutral atom	laser and microwave pulses address hyperfine or Rydberg levels

Ideal gate versus implementation

The ideal unitary is a target. Real control suffers amplitude error, detuning, leakage, crosstalk, decoherence, and calibration drift. A pulse may implement U_{error} rather than exactly U_{target} .

Virtual z rotations

A z rotation can often be implemented by changing the phase reference of subsequent control pulses rather than physically waiting for a z-axis evolution. This can make z rotations effectively fast and precise, depending on the control architecture.

27. Numerical precision and global phase

When comparing simulated matrices or statevectors, exact equality is usually inappropriate because floating-point arithmetic introduces tiny errors.

Statevector comparison up to global phase

For normalized vectors $|\psi\rangle$ and $|\phi\rangle$, they represent the same pure state when

$$|\langle\psi|\phi\rangle| \text{ approximately } 1.$$

A useful phase alignment is

$$\gamma = \arg(\langle\psi|\phi\rangle), |\phi_{\text{aligned}}\rangle = e^{-i\gamma}|\phi\rangle.$$

Matrix comparison

To compare U and V up to global phase, estimate δ from a nonzero matrix element or from

$$e^{i\delta} \text{ approximately } \text{Tr}(U^\dagger V) / |\text{Tr}(U^\dagger V)|$$

when the trace is nonzero, then compare V with $e^{i\delta}U$.

Do not discard global phase blindly in controlled circuits

A compiler may track a circuit-level global phase. It is irrelevant for unconditional measurement statistics of the complete state but can matter when moving operations into controlled branches or comparing exact operator identities.

28. Worked examples

Example 1: X acting on a general qubit

$$X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle.$$

Example 2: HZH

Using $HZH=X$, a phase flip in the Z basis becomes a bit flip in the X-transformed basis.

Example 3: prepare $(|0\rangle + i|1\rangle)/\sqrt{2}$

Apply H to $|0\rangle$ to obtain $|+\rangle$, then apply S:

$$S H|0\rangle = S|+\rangle = (|0\rangle + i|1\rangle)/\sqrt{2}.$$

Example 4: $R_y(\theta)|0\rangle$

$$R_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle.$$

Therefore a Z measurement gives

$$P(0) = \cos^2(\theta/2), P(1) = \sin^2(\theta/2).$$

Worked examples continued

Example 5: Rz does not change direct Z probabilities

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

$$R_z(\theta)|\psi\rangle = e^{-i\theta/2}\alpha|0\rangle + e^{i\theta/2}\beta|1\rangle.$$

The squared magnitudes remain $|\alpha|^2$ and $|\beta|^2$, while the relative phase increases by θ .

Example 6: noncommuting rotations

Starting from $|0\rangle$, $R_y(\pi/2)$ then $R_z(\pi/2)$ produces an equatorial state along $+y$. Reversing the order leaves R_z acting first on $|0\rangle$, where it contributes only a global phase, and R_y then produces $+x$. The outputs differ.

Example 7: arbitrary-axis pi rotation

For unit n ,

$$R_n(\pi) = -i(n \cdot \sigma).$$

Up to global phase, a π rotation about n is the Hermitian observable $n \cdot \sigma$.

Example 8: inverse circuit

If $U = R_z(a)R_y(b)R_z(c)$, then

$$U^\dagger = R_z(-c)R_y(-b)R_z(-a).$$

Reverse the order and negate every rotation angle.

29. Common mistakes

Mistake	Correction
Reading a matrix product left to right	The rightmost matrix acts first.
Treating H as a randomizer	H is a deterministic basis change; randomness depends on state and measurement basis.
Thinking Z does nothing because probabilities are real	Phases are relative phases and therefore later interference.
Forgetting half-angles in Rx, Ry, Rz	Bloch rotation theta uses $\cos(\theta/2)$ and $\sin(\theta/2)$ in the spinor matrix.
Equating $P(\lambda)$ and $R_z(\lambda)$	They differ by global phase $e^{i\lambda/2}$.
Assuming gates about different axes commute	They generally do not.
Comparing states component by component	Physical states are equivalent up to global phase.
Dropping global phase inside a controlled operation	It can operate on a relative phase between control branches.
Calling every 2×2 matrix a quantum gate	Adapted-system gate must be unitary.
Confusing gate eigenvalues with measurement probabilities	Eigenvalue predicts the gate action; probabilities depend on state overlap with eigenvectors.

A reliable workflow

For any one-qubit gate problem: write the matrix, verify the input basis ordering, multiply explicitly, simplify amplitudes, check normalization, then interpret the result using relative phase and Bloch geometry.

30. Exercises

Basic gate action

- 1 Apply X , Y , and Z to $\alpha|0\rangle + \beta|1\rangle$.
- 2 Calculate $H|+\rangle$, $H|-\rangle$, and $H[(|0\rangle + i|1\rangle)/\sqrt{2}]$.
- 3 Show that $S^2 = Z$ and $T^2 = S$.
- 4 Find the adjoints of S , T , and $P(\lambda)$.
- 5 Verify directly that $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$ are unitary.
- 6 Show that $R_x(\pi) = -iX$ and explain the role of the factor $-i$.

Composition and identities

- 1 Calculate HZH and HXH by matrix multiplication.
- 2 Calculate XY and YX .
- 3 Show that $X R_z(\theta) X = R_z(-\theta)$.
- 4 Prove $H R_z(\theta) H = R_x(\theta)$.
- 5 Starting from $|0\rangle$, compare $R_z(\pi/2)R_y(\pi/2)|0\rangle$ with $R_y(\pi/2)R_z(\pi/2)|0\rangle$.
- 6 Find the inverse of $U = R_z(a)R_y(b)R_z(c)$.

Exercises continued

State preparation and geometry

- 1 Find rotations that prepare $(\sqrt{3}/2)|0\rangle + (i/2)|1\rangle$ from $|0\rangle$.
- 2 Find θ and ϕ for the state $(1/2)|0\rangle + (\sqrt{3}/2)e^{i\pi/3}|1\rangle$.
- 3 Derive the Bloch-vector transformation produced by Z .
- 4 For $|\psi_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$, calculate probabilities after H .
- 5 Show that $R_n(\theta) = \cos(\theta/2)I - i\sin(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}$.
- 6 Explain why a 2π rotation changes a statevector sign while leaving the Bloch vector unchanged.

Advanced and conceptual

- 1 Classify I , X , H , S , T as Hermitian, unitary, Clifford, and self-inverse where applicable.
- 2 Explain why controlled- U and controlled- $(e^{i\delta}U)$ are not generally equivalent.
- 3 Show that the eigenvalues of $R_n(\theta)$ are $e^{-i\theta/2}$ and $e^{i\theta/2}$.
- 4 Describe how a basis-change circuit converts an X measurement into a Z measurement.
- 5 Explain why T is important for universal fault-tolerant computation.
- 6 Construct a circuit identity that implements $R_x(\theta)$ using H and $R_z(\theta)$.

31. Selected solutions

Exercise 2

$$H|+\rangle = |0\rangle, H|-\rangle = |1\rangle.$$

For $(|0\rangle + i|1\rangle)/\sqrt{2}$,

$$H|\psi\rangle = [(1+i)/2]|0\rangle + [(1-i)/2]|1\rangle.$$

Exercise 3

$$S^2 = \text{diag}(1, i^2) = \text{diag}(1, -1) = Z,$$

$$T^2 = \text{diag}(1, e^{i\pi/2}) = S.$$

Exercise 6

$$R_x(\pi) = \cos(\pi/2)I - i \sin(\pi/2)X = -iX.$$

The factor $-i$ is a global phase for an isolated one-qubit state.

Exercise 7

$$HZH = X, HXH = Z.$$

Exercise 8

$$XY = iZ, YX = -iZ.$$

Selected solutions continued

Exercise 9

Because $XZX = -Z$,

$$X R_z(\theta) X = X \exp(-i \theta Z/2) X = \exp(+i \theta Z/2) = R_z(-\theta).$$

Exercise 10

Using $HZH = X$,

$$H R_z(\theta) H = \exp[-i \theta (HZH)/2] = \exp(-i \theta X/2) = R_x(\theta).$$

Exercise 12

$$U^\dagger = R_z(-c) R_y(-b) R_z(-a).$$

Exercise 13

The target has $|\alpha| = \sqrt{3}/2$ and $|\beta| = 1/2$, so $\theta = \pi/3$. The relative phase is $\phi = \pi/2$. One preparation is

$$R_z(\pi/2) R_y(\pi/3) |0\rangle \text{ up to global phase.}$$

Exercise 14

$$\theta = 2 \arccos(1/2) = 2\pi/3, \phi = \pi/3.$$

Selected solutions continued

Exercise 16

$$P(0)=\cos^2(\phi/2), P(1)=\sin^2(\phi/2).$$

Exercise 17

Since $(n \cdot \sigma)^2 = I$, separate the exponential series into even and odd powers to obtain

$$R_n(\theta) = \cos(\theta/2)I - i \sin(\theta/2)n \cdot \sigma.$$

Exercise 18

The statevector acquires -1 because $R_n(2\pi) = -I$. Density matrices and Bloch vectors are unchanged because

$$(-|\psi\rangle)(-\langle\psi|) = |\psi\rangle\langle\psi|.$$

Exercise 20

The eigenstates of $n \cdot \sigma$ have eigenvalues ± 1 , so applying the exponential gives

$$\lambda_+ = e^{-i\theta/2}, \lambda_- = e^{i\theta/2}.$$

Exercise 23

$$R_x(\theta) = H R_z(\theta) H.$$

32. Compact reference sheet

Gate / concept	Formula or action
Identity	$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$; swaps alpha and beta
Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$; $[-i \beta, i \alpha]^T$
Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$; $\beta \rightarrow -\beta$
H	$(1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
$P(\lambda)$	$\text{diag}(1, e^{i\lambda})$
S	$P(\pi/2) = \text{diag}(1, i)$
T	$P(\pi/4) = \text{diag}(1, e^{i\pi/4})$
$R_x(\theta)$	$\cos(\theta/2)I - i \sin(\theta/2)X$
$R_y(\theta)$	$\cos(\theta/2)I - i \sin(\theta/2)Y$
$R_z(\theta)$	$\cos(\theta/2)I - i \sin(\theta/2)Z$
Arbitrary axis	$R_n(\theta) = \cos(\theta/2)I - i \sin(\theta/2)n \cdot \sigma$
Gate inverse	$U^{-1} = U^\dagger$
Composition	later gate multiplies on the left
General unitary	$e^{i\delta} R_z(\alpha) R_y(\beta) R_z(\gamma)$
State preparation	$R_z(\phi) R_y(\theta) 0\rangle$, up to global phase
Pauli products	$XY = iZ, YZ = iX, ZX = iY$
Hadamard conjugation	$HXH = Z, HZH = X, HYH = -Y$
Phase relation	$P(\lambda) = e^{i\lambda/2} R_z(\lambda)$
Same-axis rotations	$R_k(a) R_k(b) = R_k(a+b)$
Clifford generators	H and S
Standard universal resource	Clifford + T plus entangling gates

Final perspective

Single-qubit gates are not isolated formulas. They form a unified geometry: unitary matrices act as rigid Bloch-sphere rotations; Pauli matrices define axes and generators; Hadamard changes basis and exposes interference; phase gates control azimuth; continuous rotations synthesize arbitrary control; and Euler decompositions show that a small set of elementary rotations can express every one-qubit unitary.

The next topic is Hadamard and interference in greater algorithmic depth, followed by a dedicated Bloch-sphere volume.